

|   |                                                                                                                                                                                                                                                                                                                                 |                |     |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|
| 9 | <p>(a) e.g. zero output impedance/resistance<br/> infinite input impedance/resistance<br/> infinite (open loop) gain<br/> infinite bandwidth<br/> infinite slew rate<br/> (1 each, max. 3 )</p>                                                                                                                                 | B3             | [3] |
|   | <p>(b) (i) gain = <math>1 + (10.8 / 1.2)</math><br/> = 10</p>                                                                                                                                                                                                                                                                   | C1<br>A1       | [2] |
|   | <p>(ii) graph: straight line from (0,0) towards <math>V_{IN} = 1.0\text{ V}</math>, <math>V_{OUT} = 10\text{ V}</math><br/> horizontal line at <math>V_{OUT} = 9.0\text{ V}</math> to <math>V_{IN} = 2.0\text{ V}</math><br/> correct +9.0 V <math>\rightarrow</math> -9.0 V (and correct shape to <math>V_{IN} = 0</math>)</p> | B1<br>B1<br>B1 | [3] |